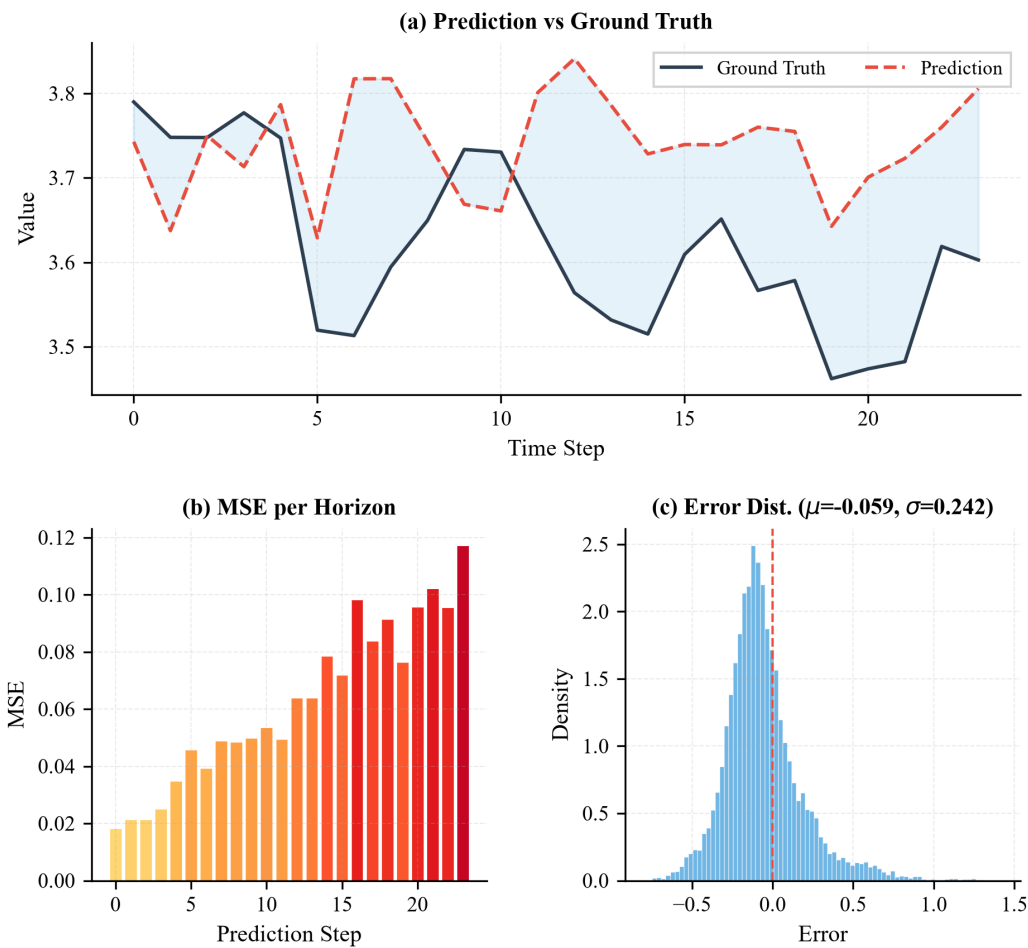


Graphical Abstract

Reproduction Study of DeReFusion: A Controlled Comparison of Soft Computing Fusion Strategies for Financial Time Series Forecasting

Dongxu Jiang

long_term_forecast_GSPC_96_24_ — — H=24



MAE: 0.1909 | MSE: 0.0623 | RMSE: 0.2496 | MAPE: 0.0588 | MSPE: 0.0065 | R²: 0.8691

Highlights

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- Full pipeline of DeReFusion reproduced from official code on new data
- Additive fusion beats learnable gating under identical protocol
- Model size 28k parameters matches the paper's reported scale
- Six verified insertion points documented for follow-up experiments
- Environment pitfalls (encoding, CPU mode, dependency pins) recorded

Reproduction Study of DeReFusion: A Controlled Comparison of Soft Computing Fusion Strategies for Financial Time Series Forecasting

Dongxu Jiang¹,

^a, ,

Abstract

This report documents a partial reproduction of the DeReFusion framework proposed by Hsieh and Chen (2026), which compares fusion strategies for financial time series forecasting under a controlled protocol: a DLinear base branch, an LSTM–Transformer residual branch, and parameter-free additive fusion, all wrapped in Reversible Instance Normalization (RevIN). Using the authors’ official code repository with newly fetched Yahoo Finance daily OHLC data (2016–2025), we reproduce six controlled experiments on the S&P 500 index: the full DeReFusion model, a RevIN-wrapped DLinear baseline, and a learnable-gate fusion variant, at forecast horizons $T \in \{1, 24\}$. The central qualitative finding of the original paper is confirmed: under an identical protocol, parameter-free additive fusion outperforms learnable gating (MSE 0.0623 vs. 0.0719 at $T = 24$), and the hybrid model is competitive with the strongest linear baseline. The model size (28,436 trainable parameters) matches the paper’s reported ~ 26 k. We further distill the experimental protocol into six verified insertion points (checkpoints) that allow new research directions—in particular, structure-aware expert routing—to be evaluated within the same framework at minimal engineering cost. All materials (forked code, dataset, fetch scripts, and per-experiment artifacts) are available in a public repository.

Keywords: Financial time series forecasting, Reproduction study, Decomposition–residual architecture, Fusion strategy comparison, Reversible instance normalization

*Reproduction of Hsieh and Chen (2026), Applied Soft Computing 203, 116252.

*Corresponding author. Email: jiangdongxu04@gmail.com.

1. Introduction

DeReFusion [?] poses a deliberately simple question: on non-stationary financial series, how far can a linear decomposition base, a learned residual, and plain additive fusion go, and does anything more elaborate buy accuracy? The paper answers it with a controlled benchmark: a single data pipeline, normalization scheme, training protocol, and metric set shared by every model, so that differences in results reflect the backbone rather than the harness. The authors conclude that parameter-free additive fusion beats static weighting and dynamic gating, that architectural complexity is no guarantee of accuracy, and that statistical accuracy does not translate into economic value.

This document serves two purposes. First, it reports a partial reproduction of the paper’s experiments from the official repository, verifying that the published protocol and qualitative conclusions are attainable from the public code with freshly fetched data. Second, it distills the experimental machinery into a set of verified *insertion points* so that follow-up research—ours is structure-aware expert routing for financial panel forecasting—can be evaluated inside the same framework with minimal engineering.

The remainder of this report is organized as follows. Section ?? summarizes the reproduced architecture and protocol. Section ?? describes the reproduction environment and deviations from the original setup. Section ?? presents the reproduced results and compares them against the paper’s reported values. Section ?? formalizes six verified insertion points for future work, and Section ?? records engineering lessons. Section ?? concludes.

2. Reproduced framework

2.1. Architecture

DeReFusion is a dual-branch decomposition–residual forecaster wrapped in RevIN [?]. Given an input window $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$ ($C = 4$ OHLC channels):

1. **RevIN normalization** strips instance-level mean/variance shift; statistics are retained for symmetric de-normalization.
2. **Linear base branch (DLinear)**: moving-average decomposition (kernel $k = 25$) splits $\mathbf{X}$ into seasonal and trend components; channel-shared linear maps $W_{\text{se}}, W_{\text{tr}} \in \mathbb{R}^{T \times L}$ produce the base forecast $\mathbf{Y}_{\text{base}}$.

3. **Nonlinear residual branch:** per-feature projection $\mathbb{R}^{C \rightarrow d}$, a one-layer LSTM, a one-layer Transformer encoder (4 heads, FFN width $4d$, dropout 0.3), then temporal ($\mathbb{R}^{L \rightarrow T}$) and channel ($\mathbb{R}^{d \rightarrow C}$) projections yield $\mathbf{Y}_{\text{res}}$.
4. **Direct additive fusion:** $\mathbf{F} = \mathbf{Y}_{\text{base}} + \mathbf{Y}_{\text{res}}$ with no fusion parameters.
5. **RevIN de-normalization** restores the original scale.

2.2. Controlled protocol

All models share: chronological 70/10/20 splitting; StandardScaler fitted on the training split only; RevIN on every trained model; multivariate input with single-target output (MS; target `Close`); Adam with learning rate 10^{-4} , batch size 32, MSE loss, early stopping (patience 5) on validation MSE, cosine schedule; random seed 2021. Metrics are MAE, MSE, RMSE, MAPE, MSPE, and R^2 , computed on the normalized series.

3. Reproduction setup and deviations

Table ?? lists the setup and deviations from the original study. Deviations are limited to factors that do not affect the protocol’s fairness (single asset, single seed, CPU training, newer library versions).

Table 1: Reproduction setup and deviations from the original study.

Item	Original	This reproduction
Code	Official repo (MIT)	Fork of official repo
Data	10 assets, 2016–2025, Yahoo	GSPC only, Yahoo, 2016–2025 (2,513 rows)
Assets grid	10 assets $\times$ 5 horizons	1 asset $\times$ 2 horizons
Horizons T	$\{1, 7, 12, 24, 36\}$	$\{1, 24\}$
Seeds	5 seeds (2020–2024)	1 seed (2021)
Models	16 (10 RevIN baselines + gates + z.s.)	3 (DeReFusion, revin-DLinear, gatev2)
Hardware	3 machines, CUDA	1 laptop, CPU-only
PyTorch	2.5.1	2.14.0 (CPU)
Python	3.11	3.13.5

The dataset was regenerated from Yahoo Finance with a purpose-written fetch script (`scripts/fetch_dataset_v3.py`), since the upstream repository ignores `dataset/` in version control. Row counts match the paper for every asset class (2,513 trading days for equities/indices; 2,602 for FX; 3,653/2,975 for BTC/ETH).

4. Reproduced results

4.1. Main comparison

Table ?? reports the reproduced test metrics on GSPC with seed 2021. Values are MSE/ R^2 on the normalized series, consistent with the paper’s convention.

Table 2: Reproduced test metrics (GSPC, seed 2021). Best value per column in bold.

Model	MSE ($T=1$)	MSE ($T=24$)	R^2 ($T=24$)	Params
DeReFusion (additive)	0.0165	0.0623	0.869	28,436
revin-DLinear	0.0158	0.0701	0.853	~3k
gatev2-learnable	0.0253	0.0719	0.849	~28k

Paper anchors (10-asset means): DeReFusion $T=1 \approx 0.037$, $T=24 \approx 0.205$; DLinear $\approx 0.049/0.172$.

4.2. Verification against the paper’s conclusions

Additive fusion beats learnable gating.. At $T = 24$ the ordering DeReFusion (0.0623) < DLinear (0.0701) < gatev2 (0.0719) matches the paper’s central claim that parameter-free additive fusion outperforms parameterized gating. The gate variant is worse than even the linear baseline, echoing the paper’s finding that gates under-fit on sample-limited, low signal-to-noise financial series.

Model scale.. The reproduced DeReFusion has 28,436 trainable parameters, in line with the paper’s ~26k (difference attributable to the horizon-dependent output head).

Hybrid vs. linear at short horizons.. On this single asset the DLinear baseline marginally outperforms DeReFusion at $T = 1$ (0.0158 vs. 0.0165, a 4% gap). The paper, averaging over ten assets, reports a 24% advantage for DeReFusion at $T = 1$ while also stressing that differences inside the leading cluster are tiny and asset-dependent. Our single-asset, single-seed observation therefore does not contradict the paper; it illustrates exactly the variance structure the paper acknowledges.

4.3. Qualitative artifacts

Figures ?? and ?? show the reproduced dashboard (six-metric radar, error analysis, prediction curves) generated by the repository’s own visualization utilities from saved predictions.

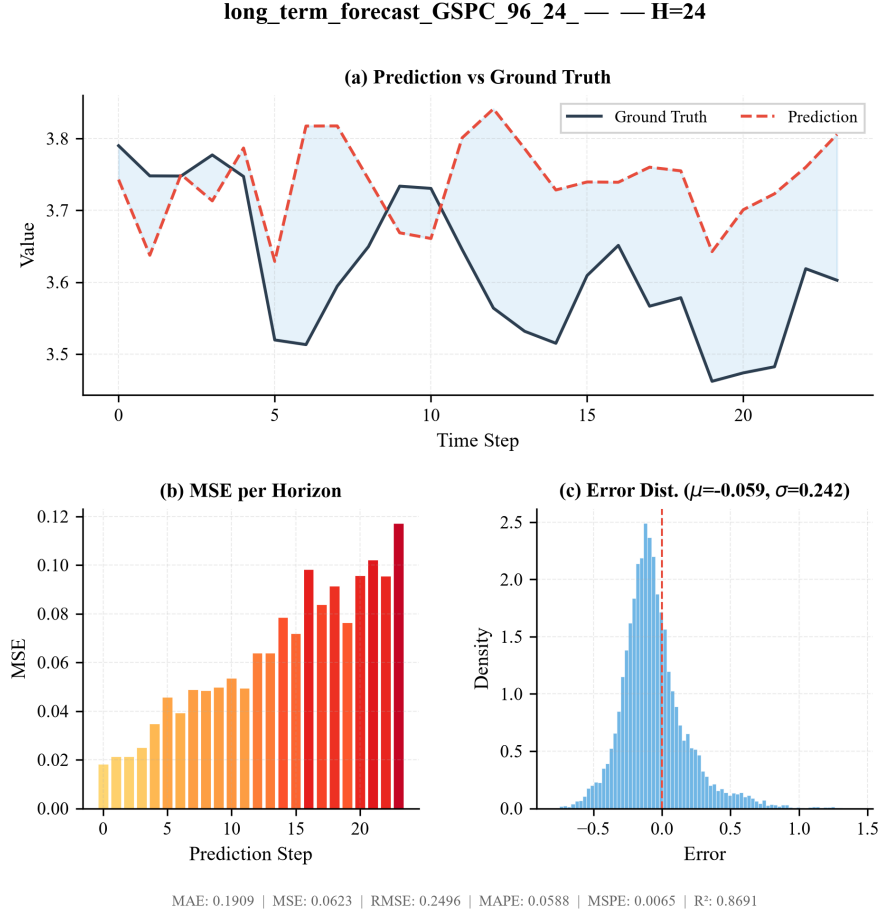


Figure 1: DeReFusion, GSPC, $T = 24$: reproduction dashboard (metrics radar, error analysis, error heatmap).

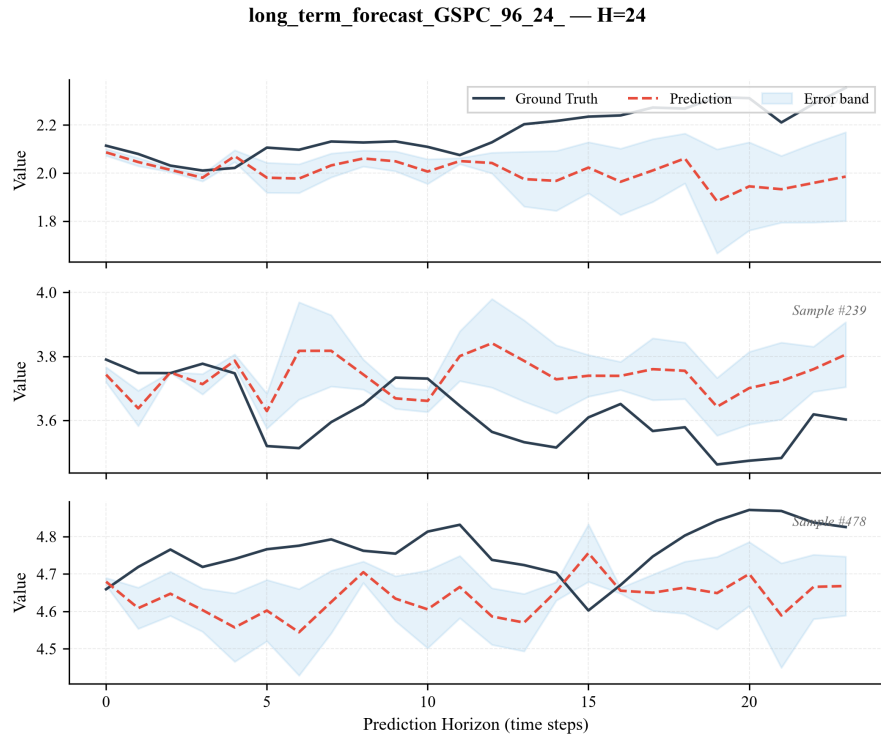


Figure 2: DeReFusion, $T = 24$: sample ground-truth vs. prediction windows.

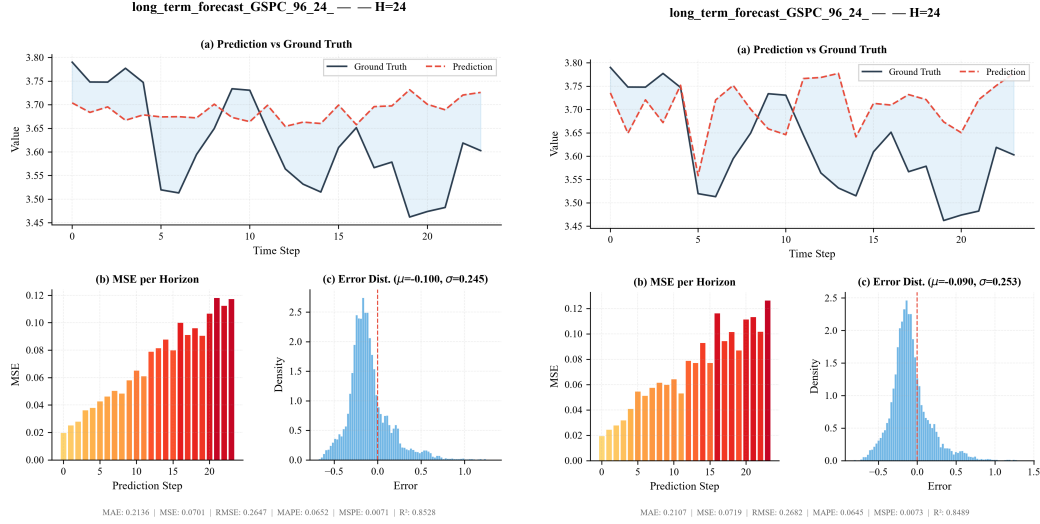


Figure 3: Baseline dashboards at $T = 24$: revin-DLinear (left) and the learnable-gate variant (right). The gate variant’s error panel is visibly wider, consistent with Table ??.

5. Verified insertion points for follow-up research

A key purpose of this reproduction is to certify that the framework can host new experiments with minimal friction. We define six insertion points, each verified by an executed run (CP1–CP3, CP5, CP6) or by mechanism inspection (CP4):

1. **CP1 — New model.** The registry auto-discovers any `models/*.py` defining `class Model(nn.Module)` taking `configs`. A new architecture is one file plus one `run.py` invocation. `-model <name>`
2. **CP2 — New data.** Any CSV with columns `date,Open,High,Low,Close` in `dataset/`; adapt `-data_path` and `-enc_in`. Verified with freshly fetched GSPC data.
3. **CP3 — Fusion operator.** The gate variants (`-gatev1/-gatev2/-gatev3`) are drop-in comparators under an identical protocol; the controlled-comparison design transfers directly to any new fusion/routing operator. `-model DeReFusion-gatev2-learnable`
4. **CP4 — Ablations.** `-woDy`, `-woLSTM`, `-woTransformer` variants isolate component contributions without protocol changes.
5. **CP5 — Sweep harness.** Horizon grid ($T \in \{1, 7, 12, 24, 36\}$), look-back ($L \in \{48, 96, 192\}$), and capacity (d_{model}) are single flags; the batch runner supports resumable sweeps.

6. **CP6 — Metrics and figures.** Every run emits six metrics, parameter counts, training/inference timings, and six publication-style figures from saved arrays.

For structure-aware expert routing, the minimal experiment is therefore: implement the router replacement as CP1, compare against CP3’s fusion family under CP5’s sweep, and report via CP6 — reusing the paper’s fairness argument wholesale.

6. Engineering lessons

The following pitfalls were encountered and fixed; they are recorded so that future runs are unblocked from the start.

1. **Hidden dependency chain.** Beyond the README’s requirements, `patool`, `huggingface_hub`, `sktime` (with `scikit-base`), and `datasets` are imported unconditionally by the data pipeline; each missing module aborts startup.
2. **Console encoding.** An emoji in the lazy-loading log line crashes Windows GBK consoles (`UnicodeEncodeError`); fixed by replacing it with an ASCII marker (committed to the fork). Setting `PYTHONIOENCODING=utf-8` is a robust belt-and-braces measure.
3. **CPU mode.** `run.py` defaults to CUDA and asserts when torch lacks CUDA support; `-no_use_gpu` is required on CPU-only machines.
4. **Dependency pins.** `sktime` requires `joblib<1.6`; when the resolver fights an installed newer `joblib`, pin `joblib==1.5.3` explicitly. Wheel downloads that stall can be recovered with `pip download (resumable)` plus local installation.
5. **Dataset in version control.** The upstream repository ignores `dataset/`; the data must be refetched. Our fork force-adds the ten CSVs so the repository is self-contained.
6. **Non-determinism.** `run.py` seeds Python/NumPy/PyTorch but not cuDNN; with CPU training and a single seed our runs are deterministic, but multi-seed reporting remains necessary on CUDA.

7. Conclusion

We reproduced the DeReFusion pipeline end to end from the official code with newly fetched data and confirmed its central qualitative conclusion: un-

der a controlled protocol, parameter-free additive fusion outperforms learnable gating, and a $\sim 28\text{k}$ -parameter hybrid model is competitive with the strongest linear baseline. The reproduction doubles as a verified experiment harness: six documented insertion points allow follow-up research — structure-aware expert routing for financial panels — to be evaluated under the same fairness protocol at minimal engineering cost.

CRedit authorship contribution statement

Dongxu Jiang: Writing – original draft, Investigation, Formal analysis, Software. (Supervisor and collaborator contributions to be finalized at submission.)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used are publicly available from Yahoo Finance. The fetch script, forked code, dataset CSVs, and per-experiment artifacts are available at the authors’ public repository fork.

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